

Automated Provisioning and Reliability Validation

Community Health Survey System

APPROACH	Ansible-managed Docker deployment
TARGET	Ubuntu 26.04.1 LTS on WSL2
STABLE RELEASE	Published image jyotsnakasi/chss-app:build-20
LOCAL CANDIDATE	Built from committed revision d7560b59fc5d
PORT	Host TCP 18084 -> container TCP 8081

Evidence date: 08 October 2026. This report uses actual Ansible, Docker, HTTP, and browser evidence. The alternate image is a local build, not a newly published Docker Hub tag.

1. Objective and target architecture

Automate CHSS deployment to a real Linux target, verify the application endpoint, demonstrate a no-change repeat run, and exercise a local candidate release followed by rollback to the known stable image.

The controller and target are the same Ubuntu 26.04.1 WSL2 distribution. Ansible uses a local inventory connection and manages the Ubuntu distro's Docker Engine; it does not target Windows Docker Desktop or its internal docker-desktop WSL distribution. The container is isolated as `chss-week14` and published only on host port 18084.

Registry discovery found build-20 as the only active published CHSS tag. To avoid inventing a release, the candidate uses the explicit local tag `local/chss-app:git-d7560b59fc5d` and carries the full source revision label. It is evaluated as a current-source build, not described as a newer published release.

2. Configuration specification

Area	Configured state	Reason / verification
Host packages	ca-certificates, docker.io, docker-buildx, python3, python3-docker	Docker Engine, Buildx candidate builds, HTTPS trust, Ansible Python runtime, Docker SDK module support
Java / Maven	Temurin Java 17 runtime; Maven 3.9.9 build stage inside the image	No host Java/Maven install is needed for the Docker deployment
Users	No additional host user; image runs as unprivileged chss	Container Dockerfile defines the runtime account
Directories	No host CHSS data/config/log directory; /app exists in image	Deployment uses an in-memory H2 database and no host bind mounts
Files	No host config files or secrets created	Runtime settings use non-secret environment values in the container
Ports	Host TCP 18084 -> container TCP 8081	Isolated Week 14 endpoint; no host MySQL port is required
Database	H2 in-memory; not MySQL	Matches the actual containerized demo deployment
Services	Ubuntu docker.service enabled and started; CHSS is a container	No host systemd CHSS or MySQL service is fabricated

3. Inventory and playbook

The inventory targets localhost with Ansible's local connection and /usr/bin/python3. The playbook gathers facts, requires a Debian-family host, refuses to replace a same-named container not labelled as Week 14-managed, installs host prerequisites, starts/enables Docker, declares the container state with community.docker.docker_container, and waits for GET /login to return HTTP 200.

```
[chss_week14]
localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3
```

The container is configured with the selected image, image/release labels, restart policy, host-to-container port mapping, and H2 environment. The playbook does not store application passwords, expose MySQL, create an unnecessary host account, or create bind-mounted data directories.

Validation commands

```
ansible-inventory -i week14_inventory.ini --graph
ansible-playbook -i week14_inventory.ini week14_provision.yml --syntax-check
ansible-playbook -i week14_inventory.ini week14_provision.yml --extra-vars chss_image=jyotsnakasi/chss-
app:build-20
```

Target environment

```
=== Ansible, Ubuntu, Docker Engine, and service verification ===
OS: Ubuntu 26.04.1 LTS
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  7.1.0
Client=29.1.3 Server=29.1.3
active
enabled
Command exit code: 0
```

4. First deployment and health verification

The first real Ansible run deployed the published build-20 image. The playbook retried its application readiness check during startup and then completed successfully. The full command output remains in the execution log; the excerpt below is selected directly from that captured output.

```
PLAY [Provision and verify the isolated CHSS Week 14 release] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Require a Debian-family Linux target] *****
ok: [localhost] => {
TASK [Inspect any existing Week 14 container without changing it] *****
ok: [localhost]
TASK [Refuse to replace a container not managed by Week 14] *****
skipping: [localhost]
TASK [Ensure host prerequisites are installed] *****
ok: [localhost]
TASK [Ensure Docker Engine is enabled and running] *****
ok: [localhost]
TASK [Deploy the requested CHSS image to the isolated Week 14 port] *****
changed: [localhost]
TASK [Wait for the CHSS login endpoint to return HTTP 200] *****
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (45 retries left).
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (44 retries left).
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (43 retries left).
ok: [localhost]
PLAY RECAP *****
localhost                : ok=7    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
Command exit code: 0
```

The explicit target-side HTTP check also returned 200. A browser opened the same login endpoint over the Ubuntu WSL2 address; the genuine screenshot shows only the blank sign-in form.

```
=== D. First HTTP health check ===
HTTP 200
Command exit code: 0
```

```
Week14-First
=== B. First provisioning and stable deployment ===
> /usr/bin/ansible-playbook -i week14_inventory.ini week14_provision.yml --extra-vars chss_image=joytsnakasi/chss-app:build-20

PLAY [Provision and verify the isolated CHSS Week 14 release] *****
TASK [Gathering Facts] *****
ok: [localhost]

TASK [Require a Debian-family Linux target] *****
ok: [localhost] => {
  "changed": false,
  "msg": "All assertions passed"
}

TASK [Inspect any existing Week 14 container without changing it] *****
ok: [localhost]

TASK [Refuse to replace a container not managed by Week 14] *****
skipping: [localhost]

TASK [Ensure host prerequisites are installed] *****
ok: [localhost]

TASK [Ensure Docker Engine is enabled and running] *****
ok: [localhost]

TASK [Deploy the requested CHSS image to the isolated Week 14 port] *****
changed: [localhost]

TASK [Wait for the CHSS login endpoint to return HTTP 200] *****
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (45 retries left).
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (44 retries left).
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (43 retries left).
ok: [localhost]

PLAY RECAP *****
localhost                : ok=7    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
Command exit code: 0
```

Actual Windows Terminal output from the first Ansible deployment.

5. Idempotency and build prerequisite

The second execution of the initial playbook reported changed=0. During source-build validation, Docker 29 reported its legacy-builder deprecation warning and the Docker CLI probe showed that Buildx was not installed. The capture helper was updated to preserve native stderr while still checking the process exit code, and the playbook now manages Ubuntu's docker-buildx package because this Week 14 workflow builds a local candidate.

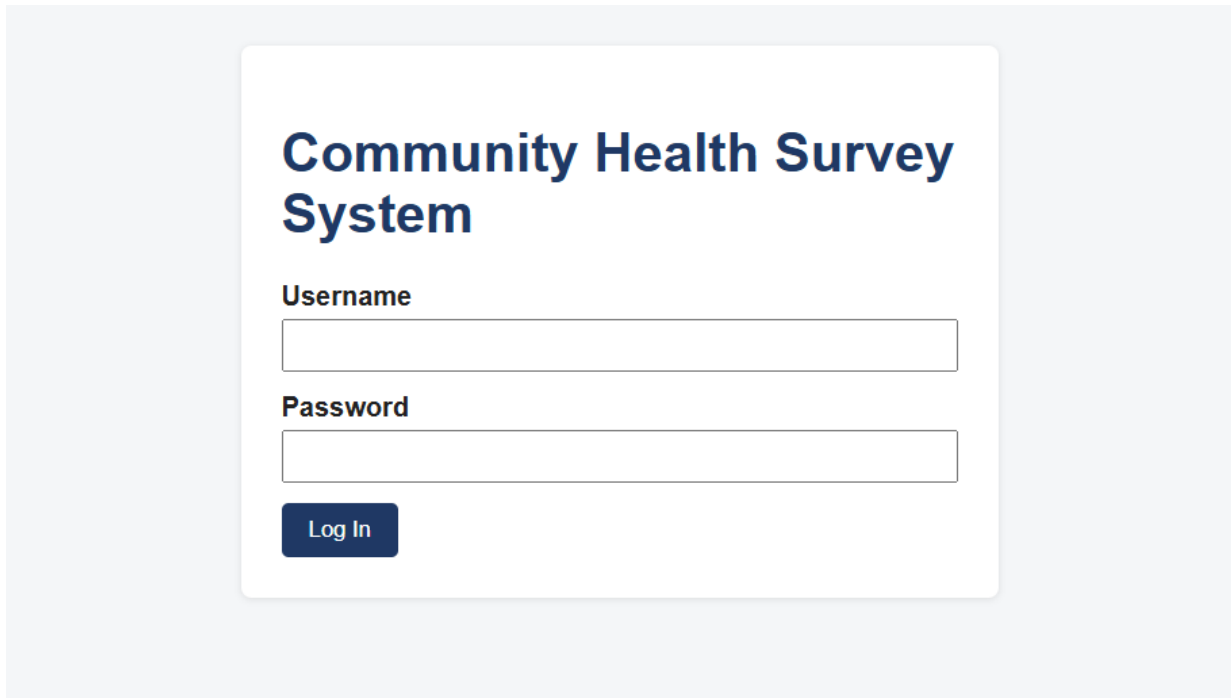
```
PLAY [Provision and verify the isolated CHSS Week 14 release] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Require a Debian-family Linux target] *****
ok: [localhost] => {
TASK [Inspect any existing Week 14 container without changing it] *****
ok: [localhost]
TASK [Refuse to replace a container not managed by Week 14] *****
ok: [localhost] => {
TASK [Ensure host prerequisites are installed] *****
ok: [localhost]
TASK [Ensure Docker Engine is enabled and running] *****
ok: [localhost]
TASK [Deploy the requested CHSS image to the isolated Week 14 port] *****
ok: [localhost]
TASK [Wait for the CHSS login endpoint to return HTTP 200] *****
ok: [localhost]
PLAY RECAP *****
localhost                : ok=8    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
ignored=0
Command exit code: 0
```

Ansible installed the missing build-only prerequisite, then a further run returned to changed=0. This final no-change run validates the current playbook after the package-list update.

```
PLAY [Provision and verify the isolated CHSS Week 14 release] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Require a Debian-family Linux target] *****
ok: [localhost] => {
TASK [Inspect any existing Week 14 container without changing it] *****
ok: [localhost]
TASK [Refuse to replace a container not managed by Week 14] *****
ok: [localhost] => {
TASK [Ensure host prerequisites and Docker build tooling are installed] *****
changed: [localhost]
TASK [Ensure Docker Engine is enabled and running] *****
ok: [localhost]
TASK [Deploy the requested CHSS image to the isolated Week 14 port] *****
ok: [localhost]
TASK [Wait for the CHSS login endpoint to return HTTP 200] *****
ok: [localhost]
PLAY RECAP *****
localhost                : ok=8    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
ignored=0
Command exit code: 0
PLAY [Provision and verify the isolated CHSS Week 14 release] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Require a Debian-family Linux target] *****
ok: [localhost] => {
TASK [Inspect any existing Week 14 container without changing it] *****
ok: [localhost]
TASK [Refuse to replace a container not managed by Week 14] *****
ok: [localhost] => {
TASK [Ensure host prerequisites and Docker build tooling are installed] *****
ok: [localhost]
TASK [Ensure Docker Engine is enabled and running] *****
ok: [localhost]
TASK [Deploy the requested CHSS image to the isolated Week 14 port] *****
ok: [localhost]
TASK [Wait for the CHSS login endpoint to return HTTP 200] *****
ok: [localhost]
PLAY RECAP *****
localhost                : ok=8    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
ignored=0
Command exit code: 0
```

6. Application response

An explicit target-side HTTP health check returned 200. A browser opened the same login endpoint over the Ubuntu WSL2 address; the genuine screenshot shows only the blank sign-in form.

A screenshot of a web browser showing the login page for the 'Community Health Survey System'. The page has a light blue background. In the center, there is a white rounded rectangle containing the title 'Community Health Survey System' in a large, bold, dark blue font. Below the title, there are two labels: 'Username' and 'Password', both in a bold, dark blue font. Under each label is a white rectangular input field with a thin grey border. At the bottom of the white rectangle is a dark blue button with the text 'Log In' in white. The entire screenshot is set against a light grey background.

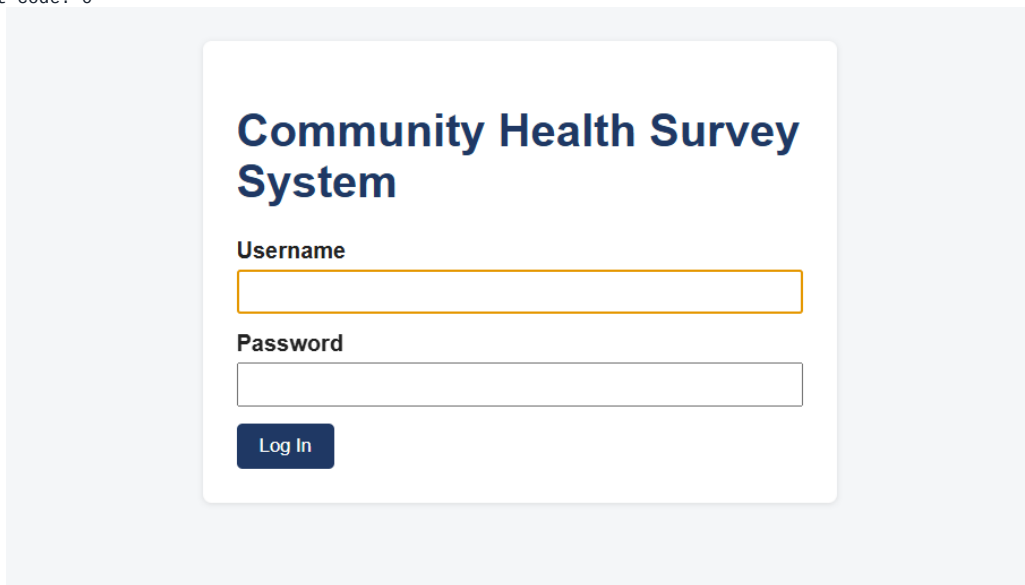
Actual browser capture of the CHSS /login page served by the deployed WSL2 container.

No test account was used, and no login credentials or seeded demo values are included in this evidence.

7. Current-source candidate and rollback

The candidate was built from a clean Git archive of committed revision d7560b59fc5df0eaa1331936c9b7a236b7f48abc. Docker labels the local image with that full revision. It is not pushed to Docker Hub. The same Ansible playbook then deploys the candidate and verifies its container identity and HTTP response.

```
github.com/docker/buildx 0.30.1 0.30.1-0ubuntu1
#15 naming to docker.io/local/chss-app:git-d7560b59fc5d
#15 naming to docker.io/local/chss-app:git-d7560b59fc5d done
#15 unpacking to docker.io/local/chss-app:git-d7560b59fc5d 0.0s done
Tags=[local/chss-app:git-d7560b59fc5d] ID=sha256:
fe893169cd4b7e31dd212725b0af72f105ea1cac81711b9580a002d74c6db72e Labels=map[org.opencontainers.image.
revision:d7560b59fc5df0eaa1331936c9b7a236b7f48abc org.opencontainers.image.version:22.04]
Command exit code: 0
Name=/chss-week14 Image=local/chss-app:git-d7560b59fc5d ImageID=sha256:
fe893169cd4b7e31dd212725b0af72f105ea1cac81711b9580a002d74c6db72e Status=running Running=true
HTTP 200
Command exit code: 0
```



Actual browser capture of the locally built candidate responding on /login.

8. Rollback and stable recovery

Rollback uses the same playbook with the previous published build-20 image. The recovery stage verifies the stable image, running state, filtered startup milestones, and HTTP endpoint response.

```
PLAY [Provision and verify the isolated CHSS Week 14 release] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Require a Debian-family Linux target] *****
ok: [localhost] => {
TASK [Inspect any existing Week 14 container without changing it] *****
ok: [localhost]
TASK [Refuse to replace a container not managed by Week 14] *****
ok: [localhost] => {
TASK [Ensure host prerequisites and Docker build tooling are installed] *****
ok: [localhost]
TASK [Ensure Docker Engine is enabled and running] *****
ok: [localhost]
TASK [Deploy the requested CHSS image to the isolated Week 14 port] *****
changed: [localhost]
TASK [Wait for the CHSS login endpoint to return HTTP 200] *****
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (45 retries left).
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (44 retries left).
FAILED - RETRYING: [localhost]: Wait for the CHSS login endpoint to return HTTP 200 (43 retries left).
ok: [localhost]
PLAY RECAP *****
localhost                : ok=8    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
ignored=0
Command exit code: 0
ID=fde9f4e3a4eb IMAGE=joytsnakasi/chss-app:build-20 STATUS=Up 32 seconds PORTS=0.0.0.0:18084->8081/tcp
NAMES=chss-week14
Name=/chss-week14 Image=joytsnakasi/chss-app:build-20 ImageID=sha256:
36a23bd53295ec2f1f0168bcc579aca7bc4b895b340985a9340991421face45e Status=running Running=true Labels={"com.
chss.application":"community-health-survey-system","com.chss.managed-by":"ansible-week14","com.chss.
release-image":"joytsnakasi/chss-app:build-20","org.opencontainers.image.version":"22.04"}
2026-10-08T09:44:29.240Z INFO 1 --- [community-health-survey] [      main] c.chss.
CommunityHealthSurveyApplication : Starting CommunityHealthSurveyApplication v0.1.0 using Java 17.0.20.1
with PID 1 (/app/app.war started by chss in /app)
2026-10-08T09:44:34.487Z INFO 1 --- [community-health-survey] [      main] o.s.b.w.embedded.tomcat.
TomcatWebServer : Tomcat started on port 8081 (http) with context path ''
2026-10-08T09:44:34.508Z INFO 1 --- [community-health-survey] [      main] c.chss.
CommunityHealthSurveyApplication : Started CommunityHealthSurveyApplication in 5.822 seconds (process
running for 6.453)
HTTP 200
Command exit code: 0
```

Community Health Survey System

Username

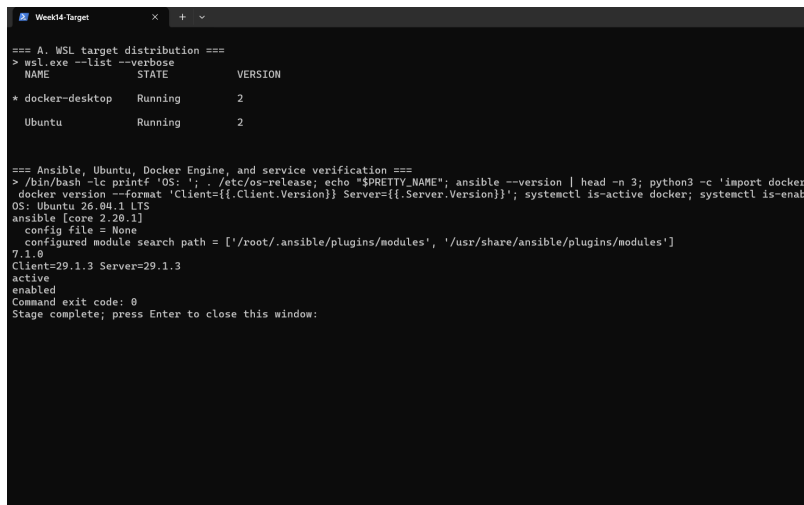
Password

Log In

Actual browser capture after rollback to the published build-20 image.

9. Terminal evidence

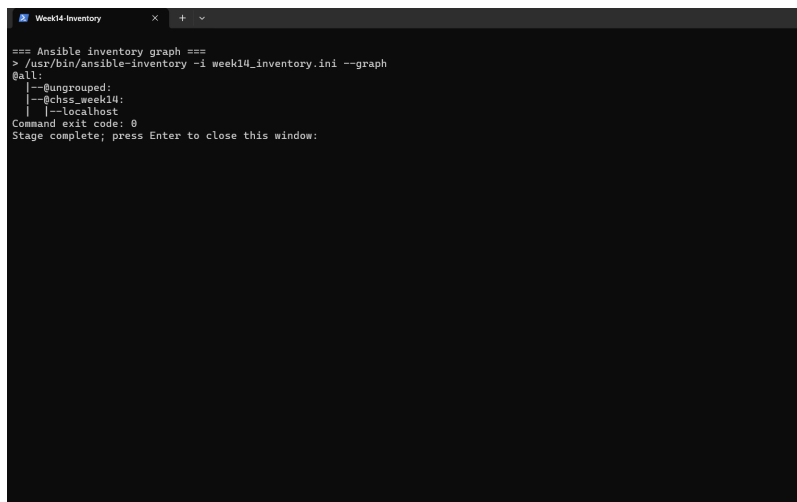
These images are genuine screen captures from the execution terminals. Each corresponds to a real command recorded in the execution log.



```
Week14-Target
=== A. WSL target distribution ===
> wsl.exe --list --verbose
  NAME      STATE      VERSION
  * docker-desktop  Running    2
  Ubuntu      Running    2

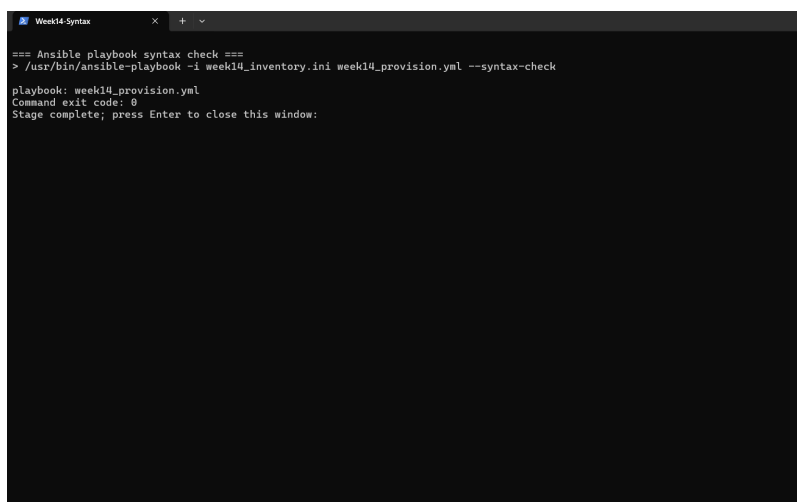
=== Ansible, Ubuntu, Docker Engine, and service verification ===
> /bin/bash -lc printf 'OS: ' ; . /etc/os-release; echo "${PRETTY_NAME}"; ansible --version | head -n 3; python3 -c 'import docker; docker version --format "Client={{.Client.Version}} Server={{.Server.Version}}"; systemctl is-active docker; systemctl is-enabled docker'
OS: Ubuntu 26.04.1 LTS
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  python interpreter = /usr/bin/python3
  ansible library path = /usr/share/ansible/libraries
Client=29.1.3 Server=29.1.3
active
enabled
Command exit code: 0
Stage complete; press Enter to close this window:
```

Ubuntu WSL2, Ansible, Python Docker SDK, Docker Engine, and service checks.



```
Week14-Inventory
=== Ansible inventory graph ===
> /usr/bin/ansible-inventory -i week14_inventory.ini --graph
@all:
  |--@ungrouped:
  | |--@chss_week14:
  | | |--localhost
Command exit code: 0
Stage complete; press Enter to close this window:
```

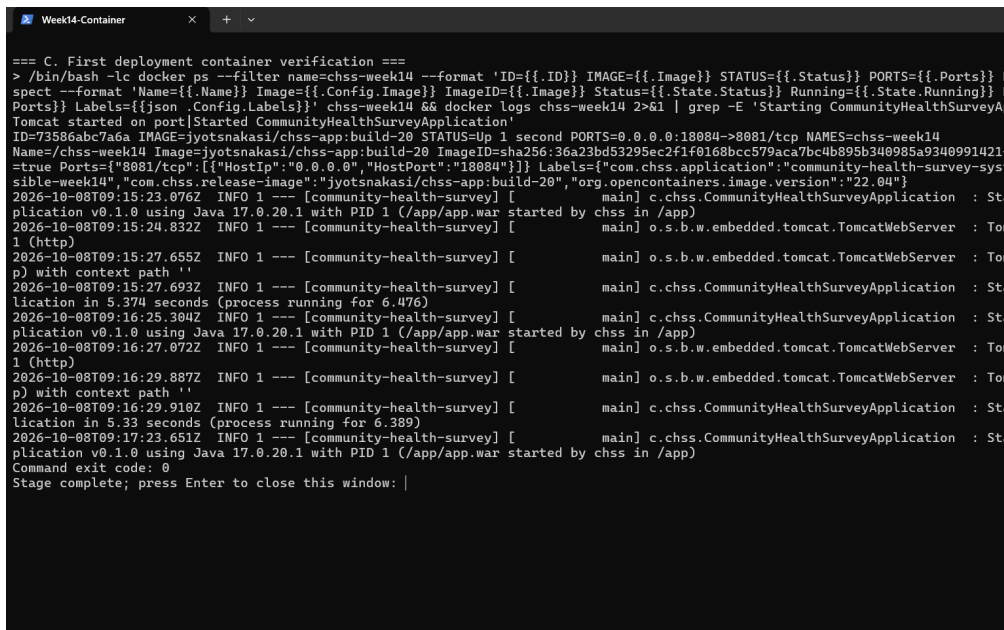
Parsed Ansible inventory graph.



```
Week14-Syntax
=== Ansible playbook syntax check ===
> /usr/bin/ansible-playbook -i week14_inventory.ini week14_provision.yml --syntax-check
playbook: week14_provision.yml
Command exit code: 0
Stage complete; press Enter to close this window:
```

Successful Ansible syntax check.

10. Deployment evidence and lessons



```

Week14-Container
=== C. First deployment container verification ===
> /bin/bash -lc docker ps --filter name=chss-week14 --format 'ID={{.ID}} IMAGE={{.Image}} STATUS={{.Status}} PORTS={{.Ports}} N
spect --format 'Name={{.Name}} Image={{.Config.Image}} ImageID={{.Image}} Status={{.State.Status}} Running={{.State.Running}} P
Ports}} Labels={{json .Config.Labels}}' chss-week14 && docker logs chss-week14 2>&1 | grep -E 'Starting CommunityHealthSurveyAp
Tomcat started on port|Started CommunityHealthSurveyApplication'
ID=73586abc7a6a IMAGE=joytsnakasi/chss-app:build-20 STATUS=Up 1 second PORTS=0.0.0.0:18080->8081/tcp NAMES=chss-week14
Names/chss-week14 Image=joytsnakasi/chss-app:build-20 ImageID=sha256:36a23bd53295ec2f1f0168bcc579ace7bcd8395b340985a9340991421f
true Ports=[{"HostIp":"0.0.0.0","HostPort":"18080"}] Labels={"com.chss.application":"community-health-survey-syst
sible-week14","com.chss.release-image":"joytsnakasi/chss-app:build-20","org.opencontainers.image.version":"22.00"}
2026-10-08T09:15:23.076Z INFO 1 --- [community-health-survey] [ main] c.chss.CommunityHealthSurveyApplication : Sta
plication v0.1.0 using Java 17.0.20.1 with PID 1 (/app/app.war started by chss in /app)
2026-10-08T09:15:24.832Z INFO 1 --- [community-health-survey] [ main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tom
1 (http)
2026-10-08T09:15:27.655Z INFO 1 --- [community-health-survey] [ main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tom
p) with context path ''
2026-10-08T09:15:27.693Z INFO 1 --- [community-health-survey] [ main] c.chss.CommunityHealthSurveyApplication : Sta
lication in 5.374 seconds (process running for 6.476)
2026-10-08T09:16:25.304Z INFO 1 --- [community-health-survey] [ main] c.chss.CommunityHealthSurveyApplication : Sta
plication v0.1.0 using Java 17.0.20.1 with PID 1 (/app/app.war started by chss in /app)
2026-10-08T09:16:27.072Z INFO 1 --- [community-health-survey] [ main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tom
1 (http)
2026-10-08T09:16:29.887Z INFO 1 --- [community-health-survey] [ main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tom
p) with context path ''
2026-10-08T09:16:29.910Z INFO 1 --- [community-health-survey] [ main] c.chss.CommunityHealthSurveyApplication : Sta
lication in 5.33 seconds (process running for 6.389)
2026-10-08T09:17:23.651Z INFO 1 --- [community-health-survey] [ main] c.chss.CommunityHealthSurveyApplication : Sta
plication v0.1.0 using Java 17.0.20.1 with PID 1 (/app/app.war started by chss in /app)
Command exit code: 0
Stage complete; press Enter to close this window: |

```

Container inspection and filtered startup milestones. Long terminal lines are also preserved as text in the execution log.

Problems encountered and resolution

Two initial capture-helper errors occurred before provisioning: one used an unsupported PowerShell output parameter, and one contained an invalid text-normalization edit. Both failures are recorded in the execution log and preserved as original error screenshots. The helper was corrected before the successful runs; neither failed attempt changed the application target.

The WSL Docker service also received orderly stop signals between short command sessions, which interrupted the container and caused connection resets during early inspection. Journal output showed graceful Docker shutdown rather than an application crash or OOM event. The Ubuntu WSL target was kept available while the subsequent deployment, health, and lifecycle checks were performed; the final HTTP checks and recovery results are recorded below.

Limitations

Docker Hub did not expose a newer CHSS tag during registry discovery. The alternate image is therefore a local, commit-labelled build, not a released or registry-published upgrade. The Dockerfile build uses -DskipTests, so this workflow verifies deployment and HTTP health but does not claim a Maven or Selenium test-suite pass. The Ubuntu target is WSL2 on a development workstation rather than a separate production server. Docker Desktop and unrelated Windows containers were not changed.

11. Final verification

Inventory parsing, Ansible syntax validation, first deployment, second-run idempotency, candidate response, rollback, and stable recovery are all represented by actual command output in the execution log. The browser screenshot confirms that the login page renders; the HTTP checks confirm the endpoint status without submitting credentials.

```
=== J. Verify stable recovery, startup logs, and endpoint ===
ID=fde9f4e3a4eb IMAGE=joytsnakasi/chss-app:build-20 STATUS=Up 32 seconds PORTS=0.0.0:18084->8081/tcp
NAMES=chss-week14
Name=/chss-week14 Image=joytsnakasi/chss-app:build-20 ImageID=sha256:
36a23bd53295ec2f1f0168bcc579aca7bc4b895b340985a9340991421face45e Status=running Running=true Labels={"com.
chss.application":"community-health-survey-system", "com.chss.managed-by":"ansible-week14", "com.chss.
release-image":"joytsnakasi/chss-app:build-20", "org.opencontainers.image.version":"22.04"}
2026-10-08T09:44:29.240Z INFO 1 --- [community-health-survey] [      main] c.chss.
CommunityHealthSurveyApplication : Starting CommunityHealthSurveyApplication v0.1.0 using Java 17.0.20.1
with PID 1 (/app/app.war started by chss in /app)
2026-10-08T09:44:30.984Z INFO 1 --- [community-health-survey] [      main] o.s.b.w.embedded.tomcat.
TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-10-08T09:44:34.487Z INFO 1 --- [community-health-survey] [      main] o.s.b.w.embedded.tomcat.
TomcatWebServer : Tomcat started on port 8081 (http) with context path ''
2026-10-08T09:44:34.508Z INFO 1 --- [community-health-survey] [      main] c.chss.
CommunityHealthSurveyApplication : Started CommunityHealthSurveyApplication in 5.822 seconds (process
running for 6.453)
HTTP 200
Command exit code: 0
```

The execution log is the authoritative transcript for command details and return codes. The PDF includes selected output and genuine screenshots for readable evidence. No secrets, passwords, access tokens, or personal data are intentionally included.

12. Conclusion

Week 14 demonstrates an Ansible-managed container release, real HTTP verification, repeat-run idempotency, deployment of a local current-source candidate, and recovery to the published stable image. All release claims are bounded by the registry evidence: build-20 is the published version, while the candidate remains local.